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A	 PT. ABB Sakti Industri Jalan Gajah Tunggal km 1.0 Tangerang 15135, Indonesia								A																																																				
B									B																																																				
C	Project Name : MV Switchgear for MNA Serang Plant Switchgear Name : MVD-29 & MVD-30 20kV Centralized Panel Room WICA Panel Type : UNISEC Customer Name : PT Multimas Nabati Asahan Contract No. : BE318042 TITLE : SLD & GA Drawing								C																																																				
D									D																																																				
E									E																																																				
F	<table><tr><td></td><td></td><td></td><td></td><td>Date Prep.</td><td>15.11.2018</td><td>PROJECT TITLE</td><td>ABB</td><td>EPMV</td><td>COVER SHEET</td><td>Scale</td><td></td><td>= H00</td></tr><tr><td>1</td><td>Issued for Approval</td><td>18.12.2018</td><td>YR</td><td>Prepared</td><td>YR</td><td>MV Switchgear for MNA Serang Plant</td><td colspan="2"></td><td>SLD & GA Drawing</td><td>Customer</td><td>PT Multimas Nabati Asahan</td><td>& GAD</td></tr><tr><td>0</td><td>First Issued</td><td>19.11.2018</td><td>YR</td><td>Checked</td><td>AW</td><td></td><td colspan="2">PT. ABB Sakti Industri</td><td>MVD-29 & MVD-30 20kV Centralized Panel Room WICA</td><td>Drawing Nbr.</td><td></td><td>Sh. 1</td></tr><tr><td>Ind.</td><td>Revision</td><td>Date</td><td>Name</td><td>Approved</td><td>AI</td><td>Based on:</td><td colspan="2">BE318042</td><td></td><td>Customer Doc. Nbr.</td><td></td><td>Cont. 23</td></tr></table>												Date Prep.	15.11.2018	PROJECT TITLE	ABB	EPMV	COVER SHEET	Scale		= H00	1	Issued for Approval	18.12.2018	YR	Prepared	YR	MV Switchgear for MNA Serang Plant			SLD & GA Drawing	Customer	PT Multimas Nabati Asahan	& GAD	0	First Issued	19.11.2018	YR	Checked	AW		PT. ABB Sakti Industri		MVD-29 & MVD-30 20kV Centralized Panel Room WICA	Drawing Nbr.		Sh. 1	Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318042			Customer Doc. Nbr.		Cont. 23	F
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B	PAGE		PAGE NAME		REVISION								
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	==MVD-29_MVD-30=H00&GAD/2		INDEX OF SHEETS		0	1							
	==MVD-29_MVD-30=H00&GAD/3		GENERAL CHARACTERISTICS		0	1							
	==MVD-29_MVD-30=H00&GAD/4		REFERENCE DESIGNATIONS		0	1							
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	==MVD-29_MVD-30=H00&GAD/6		REFERENCE DESIGNATIONS		0	1							
	==MVD-29_MVD-30=H00&GAD/7		REFERENCE DESIGNATIONS		0	1							
	==MVD-29_MVD-30=H00&GAD/8		SINGLE LINE DIAGRAM		0	1							
	==MVD-29_MVD-30=H00&GAD/9		SINGLE LINE DIAGRAM		0	1							
	==MVD-29_MVD-30=H00&GAD/10		SINGLE LINE DIAGRAM		0	1							
	==MVD-29_MVD-30=H00&GAD/11		INSTALLATION RULES		0	1							
	==MVD-29_MVD-30=H00&GAD/12		FRONT VIEW & GA		0	1							
	==MVD-29_MVD-30=H00&GAD/13		FRONT VIEW & GA		0	1							
	==MVD-29_MVD-30=H00&GAD/14		FOUNDATION FRAMES		0	1							
	==MVD-29_MVD-30=H00&GAD/15		FOUNDATION FRAMES		0	1							
	==MVD-29_MVD-30=H00&GAD/16		FOUNDATION FRAMES DETAILS		0	1							
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	==MVD-29_MVD-30=H00&GAD/21		SECTION VIEW		0	1							
	==MVD-29_MVD-30=H00&GAD/22		SECTION VIEW		0	1							
	==MVD-29_MVD-30=H00&GAD/23		SECTION VIEW		0	1							
D									D				
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A	CHARACTERISTICS (IN COMPLIANCE WITH STANDARD IEC 62271-200)				NOTES				A
B	SWITCHGEAR VERSION = UNISEC				COLOUR OF FRONT DOORS = RAL 7035				B
	RATED VOLTAGE (Un) = 24 kV				SIDEWALLS PAINTED = YES				
C	OPERATING VOLTAGE = 20 kV				SIDES = L+R				C
	RATED FREQUENCY (fr) = 50 Hz								
D	RATED LIGHTNING IMPULSE WITHSTAND VOLTAGE (Up) = 125 kV				SUPPLIED IMPLEMENTS				D
	RATED POWER FREQUENCY WITHSTAND VOLTAGE (Ud) = 50 kV								
E	RATED CURRENT OF MAIN BUSBARS (Ir) = 630 A				IDENTIFICATION OF CONDUCTORS				E
	RATED SHORT-TIME WITHSTAND CURRENT (Ik) = 20 kA								
F	RATED PEAK WITHSTAND CURRENT (Ip) = 50 kA				FOR IDENTIFICATION OF CONDUCTORS THE "LOCAL END CONNECTION LABELLING" SYSTEM IS USED, IN COMPLIANCE WITH STANDARD IEC 62491 PARAGRAPH 6.2. THIS SYSTEM FORESEES THAT THE MARKING OF A CONDUCTOR END SHOWS THE DESIGNATION OF THE TERMINAL TO WHICH IT IS CONNECTED. IN COMPLIANCE WITH STANDARD IEC 61666 EACH CONDUCTOR IS IDENTIFIED WITH THE REFERENCE DESIGNATION OF THE CONNECTED ELECTRICAL COMPONENT FOLLOWED BY SIGN ":" (COLON), FOLLOWED BY THE TERMINAL DESIGNATION. FOR EXAMPLE THE CONDUCTOR CONNECTED TO TERMINAL "10" OF THE COMPONENT WITH DESIGNATION "-XA" HAS THE MARKING "-XA:10". THE CONDUCTOR MARKING IS PLACED IN SUCH A WAY TO READ IT ALWAYS FROM THE TERMINAL TOWARDS THE CONDUCTOR, AS SHOWN BELOW -XDA 9 10 11 -XDA:10 -XDA:10				F
	RATED DURATION OF SHORT CIRCUIT (tk) = 1 s								
A	INTERNAL ARC CLASSIFICATION (IAC) = AFLR / 16kAx1s				IDENTIFICATION OF CONDUCTORS				A
	AMBIENT CONDITION = NORMAL								
B	AMBIENT AIR TEMPERATURE = -5°C...+40°C				IDENTIFICATION OF CONDUCTORS				B
	DEGREE OF PROTECTION (OPERATION SEATS EXCLUDED) = IP3X								
C	DEGREE OF PROTECTION WITH OPEN DOORS = IP2X				IDENTIFICATION OF CONDUCTORS				C
	RATED SUPPLY VOLTAGE OF CONTROL AND SIGNALLING CIRCUITS (Ua) = 230VAC								
D	RATED SUPPLY VOLTAGE OF SPRING CHARGING MOTOR (Ua) = 230VAC				IDENTIFICATION OF CONDUCTORS				D
	RATED SUPPLY VOLTAGE OF LIGHTING AND HEATING CIRCUITS (Ua) = 230VAC								
E	TYPE OF CABLE FOR LOW VOLTAGE CONDUCTORS = 450/750V				IDENTIFICATION OF CONDUCTORS				E
	TYPE OF CABLE FOR LOW VOLTAGE CONDUCTORS = PVC								
F	CROSS-SECTION OF CONDUCTORS FOR CURRENT CIRCUITS = 2.5 mm²/BLACK				IDENTIFICATION OF CONDUCTORS				F
	CROSS-SECTION OF CONDUCTORS FOR VOLTAGE CIRCUITS = 1.5 mm²/BLACK								
A	CROSS-SECTION OF CONDUCTORS FOR MOTORS = 2.5 mm²/BLACK				IDENTIFICATION OF CONDUCTORS				A
	CROSS-SECTION OF OTHER CONDUCTORS (CONTROL AND SIGNALLING CIRCUITS) = 1.5 mm²/BLACK								
B	CROSS-SECTION OF CONDUCTORS FOR INTERCONNECTIONS = 1.5 mm²/BLACK				IDENTIFICATION OF CONDUCTORS				B
	CROSS-SECTION OF CONDUCTORS FOR INTERCONNECTIONS OF SUPPLY VOLTAGE = 2.5 mm²/BLACK								
C	COMMUNICATION CABLE = N/A				IDENTIFICATION OF CONDUCTORS				C
D					IDENTIFICATION OF CONDUCTORS				D
E					IDENTIFICATION OF CONDUCTORS				E
F					IDENTIFICATION OF CONDUCTORS				F

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A	REFERENCE DESIGNATION OF OBJECTS IN ELECTRICAL DOCUMENTS																			
	(IN COMPLIANCE WITH STANDARD IEC 81346-2 AND ABB TECHNICAL STANDARD 2NBA000001)																			
	DESIGNATION	DESCRIPTION																		
	-AA	MULTIFUNCTION UNIT (CENTRAL UNIT)																		
	-BAD1,-BAD4	CAPACITIVE VOLTAGE DIVIDER LOCATED ON PHASE L1																		
	-BAD2,-BAD5	CAPACITIVE VOLTAGE DIVIDER LOCATED ON PHASE L2																		
	-BAD3,-BAD6	CAPACITIVE VOLTAGE DIVIDER LOCATED ON PHASE L3																		
B	-BAR	VOLTAGE PROTECTION RELAY																		
	-BAS1	VOLTAGE SENSOR LOCATED ON PHASE L1																		
	-BAS2	VOLTAGE SENSOR LOCATED ON PHASE L2																		
	-BAS3	VOLTAGE SENSOR LOCATED ON PHASE L3																		
	-BAT1	VOLTAGE TRANSFORMER LOCATED ON PHASE L1																		
	-BAT2	VOLTAGE TRANSFORMER LOCATED ON PHASE L2																		
	-BAT3	VOLTAGE TRANSFORMER LOCATED ON PHASE L3																		
	-BAT4	VOLTAGE TRANSFORMER IN BUSBAR COMPARTMENT LOCATED ON PHASE L1																		
	-BAT5	VOLTAGE TRANSFORMER IN BUSBAR COMPARTMENT LOCATED ON PHASE L2																		
	-BAT6	VOLTAGE TRANSFORMER IN BUSBAR COMPARTMENT LOCATED ON PHASE L3																		
C	-BCD	DIFFERENTIAL PROTECTIVE RELAY																		
	-BCF	FEEDER PROTECTION RELAY																		
	-BCG	GENERATOR PROTECTION RELAY																		
	-BCM	MOTOR PROTECTION RELAY																		
	-BCN	NEUTRAL (RESIDUAL) CURRENT TRANSFORMER																		
	-BCP	TRANSFORMER PROTECTION RELAY																		
	-BCR	CURRENT PROTECTION RELAY																		
	-BCS1	CURRENT SENSOR LOCATED ON PHASE L1																		
	-BCS2	CURRENT SENSOR LOCATED ON PHASE L2																		
	-BCS3	CURRENT SENSOR LOCATED ON PHASE L3																		
D	-BCT1	CURRENT TRANSFORMER LOCATED ON PHASE L1																		
	-BCT2	CURRENT TRANSFORMER LOCATED ON PHASE L2																		
	-BCT3	CURRENT TRANSFORMER LOCATED ON PHASE L3																		
	-BCT4,-BCT7	CURRENT TRANSFORMER LOCATED ON PHASE L1																		
	-BCT5,-BCT8	CURRENT TRANSFORMER LOCATED ON PHASE L2																		
	-BCT6,-BCT9	CURRENT TRANSFORMER LOCATED ON PHASE L3																		
	-BCZ	DISTANCE PROTECTION RELAY																		
	-BEF	FREQUENCY PROTECTION RELAY																		
	-BER	SUPERVISION RELAYS																		
	-BES	SYNCHRONIZING RELAY																		
	-BET	THERMAL PROTECTION RELAY																		
E	-BGA1	POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN CIRCUIT-BREAKER COMPARTMENT																		
	-BGA2	POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN BUSBAR COMPARTMENT (SINGLE BUSBAR SYSTEM) OR IN FRONT BUSBAR COMPARTMENT (DOUBLE BUSBAR SYSTEM)																		
	-BGA3	POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN CABLE COMPARTMENT																		
	-BGA4	POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN REAR BUSBAR COMPARTMENT (DOUBLE BUSBAR SYSTEM)																		
	-BGB1...-BGB3	POSITION SWITCHES OF CIRCUIT-BREAKER																		
	-BGB4	POSITION SWITCH OF CIRCUIT-BREAKER: PASSING CONTACT CLOSING MOMENTARILY WHEN THE BREAKER OPENS																		
	-BGB5	POSITION SWITCH OF CIRCUIT-BREAKER SIGNALLING UNDERVOLTAGE RELEASE ENERGIZED																		
	-BGB6	POSITION SWITCH OF CIRCUIT-BREAKER SIGNALLING UNDERVOLTAGE RELEASE EXCLUDED MECHANICALLY																		
	-BGB7	POSITION SWITCHES OF CIRCUIT-BREAKER ON SWITCHGEAR																		
	-BGB8...-BGB10	POSITION SWITCHES OF CIRCUIT-BREAKER																		
	-BGB11	POSITION SWITCH OF CIRCUIT-BREAKER OPERATING FOR MANUAL OPENING																		
	-BGB12	PROXIMITY SWITCH SIGNALLING CIRCUIT-BREAKER IN OPEN POSITION																		
	-BGB13	PROXIMITY SWITCH SIGNALLING CIRCUIT-BREAKER IN CLOSED POSITION																		
	-BGD	POSITION SWITCH OF L.V. INSTRUMENT COMPARTMENT DOOR.																		
	-BGD1	POSITION SWITCH OF CIRCUIT-BREAKER COMPARTMENT DOOR																		
	-BGD2	POSITION SWITCH OF CABLE COMPARTMENT DOOR																		
	-BGE1	POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE IN OPEN POSITION																		
	-BGE2	POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE IN CLOSED POSITION																		
	-BGE3	POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)																		
	-BGE4	POSITION SWITCHES OF OPERATION OF EARTHING SWITCH -QCE ACTUATED BY ELECTRIC MOTOR																		
	-BGE5	POSITION SWITCHES OF OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2 ACTUATED BY ELECTRIC MOTOR																		
	-BGE7	POSITION SWITCH SIGNALLING EARTHING SWITCH NOT IN MANUAL OPERATION																		
	-BGE8	POSITION SWITCHES OF EARTHING SWITCH -QCE SUPPLIED TOGETHER WITH SWITCH- DISCONNECTOR -QBS																		
	-BGE11	POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE1 OR -QCE2 IN OPEN POSITION																		
	-BGE12	POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE1 OR -QCE2 IN CLOSED POSITION																		
	-BGE13	POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE1 OR -QCE2 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)																		
	-BGE14, -BGE15	POSITION SWITCHES FOR OPERATION OF EARTHING SWITCH -QCE SUPPLIED TOGETHER WITH SWITCH-DISCONNECTOR -QBS, ACTUATED BY ELECTRIC MOTOR																		
	-BGF	POSITION SWITCHES OF MEDIUM VOLTAGE FUSES																		
	-BG11	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD IN OPEN POSITION																		
	-BG12	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD IN CLOSED POSITION																		
	-BG13	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)																		
	-BG14	POSITION SWITCHES OF OPERATION OF DISCONNECTOR -QBD1 ACTUATED BY ELECTRIC MOTOR																		
	-BG15	POSITION SWITCHES OF OPERATION OF DISCONNECTOR -QBD2 ACTUATED BY ELECTRIC MOTOR																		
	-BG16	POSITION SWITCH SIGNALLING DISCONNECTOR (OR SWITCH-DISCONNECTOR) -QBD NOT IN EMERGENCY MANUAL OPERATION																		
	-BG17	POSITION SWITCH SIGNALLING DICONNECTOR (OR SWITCH-DISCONNECTOR) -QBD NOT IN MANUAL OPERATION																		
	-BG18	POSITION SWITCHES OF SWITCH-DISCONNECTOR -QBS																		
	-BG19	POSITION SWITCHES OF OPERATION OF SWITCH-DISCONNECTOR -QBS ACTUATED BY ELECTRIC MOTOR																		
	-BG111	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD1 IN OPEN POSITION																		
	-BG112	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD1 IN CLOSED POSITION																		
	-BG113	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD1 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)																		
	-BG114, -BG115	POSITION SWITCHES OF OPERATION OF DISCONNECTOR -QBD1 ACTUATED BY ELECTRIC MOTOR																		
	-BG121	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD2 IN OPEN POSITION																		
	-BG122	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD2 IN CLOSED POSITION																		
	-BG123	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD2 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)																		
	-BG124, -BG125	POSITION SWITCHES OF OPERATION OF DISCONNECTOR -QBD2 ACTUATED BY ELECTRIC MOTOR																		
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A	-BGI31		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION		-CA		CAPACITORS		A						
	-BGI32		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN CLOSED POSITION		-CA1		CAPACITOR FOR FRONT VENTILATOR								
	-BGI33		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)		-CA2		CAPACITOR FOR REAR VENTILATOR								
	-BGI41...		POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN FEEDER POSITION		-EA1		LIGHTING LAMP LOCATED IN L.V. INSTRUMENT COMPARTMENT.								
	-BGI42		POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS IN OPEN POSITION		-EA2		LIGHTING LAMP LOCATED IN CABLE COMPARTMENT								
	-BGI43...		POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN EARTH POSITION		-EA3		LIGHTING LAMP LOCATED IN OPERATING MECHANISM ENCLOSURE								
	-BGI51		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN FEEDER POSITION		-EA4		LIGHTING LAMP LOCATED IN CIRCUIT BREAKER COMPARTIMENT								
	-BGI52		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 IN OPEN POSITION		-EB1		HEATER LOCATED IN CABLE COMPARTMENT								
	-BGI53		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN EARTH POSITION		-EB2		HEATER LOCATED IN CIRCUIT-BREAKER COMPARTMENT								
	-BGK		POSITION SWITCH OF KEY LOCK		-EB3		HEATER LOCATED IN MOTOR CABINET								
B	-BGL1		POSITION SWITCH OF ELECTROMECHANICAL LOCK -RLE1		-EB4		HEATER FOR VENTILATION TEST		B						
	-BGS1....BGS4		POSITION SWITCHES OF CIRCUIT-BREAKER SPRINGS		-EB5		HEATER LOCATED IN L.V. INSTRUMET COMPARTMENT.								
	-BGS5		PROXIMITY SWITCH SIGNALLING SPRINGS IN CHARGED POSITION		-EB7		HEATERS LOCATED IN LEFT SIDE OF THE OPERATING MECHANISM ENCLOSURE								
	-BGT1		POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN SERVICE POSITION		-EB8		HEATERS LOCATED IN RIGHT SIDE OF THE OPERATING MECHANISM ENCLOSURE								
	-BGT2		POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN TEST POSITION		-FA1		SURGE ARRESTER LOCATED ON PHASE L1								
	-BGT3		POSITION SWITCH ON TRUCK SIGNALLING TRUCK NOT IN ISOLATING TRAVEL POSITION		-FA2		SURGE ARRESTER LOCATED ON PHASE L2								
	-BGT4		POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN SERVICE POSITION		-FA3		SURGE ARRESTER LOCATED ON PHASE L3								
	-BGT5		POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN TEST POSITION		-FCD		FUSE-DISCONNECTORS FOR PROTECTION OF AUXILIARY CIRCUITS								
	-BGT6		POSITION SWITCHES SIGNALLING V.T. TRUCK IN SERVICE POSITION		-FCF1		MEDIUM VOLTAGE FUSE LOCATED ON PHASE L1								
	-BGT7		POSITION SWITCHES SIGNALLING V.T. TRUCK IN TEST POSITION		-FCF2		MEDIUM VOLTAGE FUSE LOCATED ON PHASE L2								
C	-BGT8		POSITION SWITCHES SIGNALLING EARTHING TRUCK IN SERVICE POSITION		-FCF3		MEDIUM VOLTAGE FUSE LOCATED ON PHASE L3		C						
	-BGT11		PROXIMITY SWITCH SIGNALLING TRUCK IN SERVICE POSITION		-FCM1		MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF SPRINGS CHARGING MOTOR ON MAIN CIRCUIT-BREAKER								
	-BGT12		PROXIMITY SWITCH SIGNALLING TRUCK IN TEST POSITION		-FCM2...-FCM10		MINIATURE CIRCUIT-BREAKERS FOR PROTECTION OF AUXILIARY CIRCUITS								
	-BGV1		POSITION SWITCH OF FRONT VENTILATOR		-FCM11		MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV1								
	-BGV2		POSITION SWITCH OF REAR VENTILATOR		-FCM12		MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV2								
	-BM		HYGROSTAT		-FCM13		MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV3								
	-BPA4		PRESSURE SWITCH LOCATED IN CABLE COMPARTMENT FOR DETECTION OF INTERNAL ARCING		-FCT		THERMAL OVERLOAD RELEASES								
	-BPA5		PRESSURE SWITCH LOCATED IN CIRCUIT-BREAKER OR IN UPPER VT COMPARTMENT FOR DETECTION OF INTERNAL ARCING		-GA		GENERATORS								
	-BPA6		PRESSURE SWITCH LOCATED IN BUSBAR OR IN BUSBAR AND CIRCUIT-BREAKER COMPARTMENT FOR DETECTION OF INTERNAL ARCING		-GQ1		FRONT VENTILATOR								
	-BPA7		PRESSURE SWITCH LOCATED IN VOLTAGE TRANSFORMER ISOLATING COMPARTMENT FOR DETECTION OF INTERNAL ARCING		-GQ2		REAR VENTILATOR								
D	-BPS1		PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER OR ON POLE OF PHASE L1 OF CIRCUIT-BREAKER		-GQ3		VENTILATOR LOCATED ON CIRCUIT-BREAKER TRUCK		D						
	-BPS2		PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L2		-KFA1		AUXILIARY RELAY SIGNALLING LOW GAS PRESSURE								
	-BPS3		PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L3		-KFA2		AUXILIARY RELAY SIGNALLING INSUFFICIENT GAS PRESSURE								
	-BR		FLAME DETECTORS, SMOKE DETECTORS		-KFA3....KFA9		AUXILIARY RELAYS OR CONTACTORS								
	-BT		THERMOSTAT		-KFC		CLOSING RELAYS OR CONTACTORS								
	-BT1		THERMOSTAT FOR DE-ENERGIZATION OF VENTILATORS		-KFI		INTEGRATED CIRCUITS								
	-BT2		THERMOSTAT FOR ENERGIZATION OF VENTILATORS		-KFL		LOCKOUT RELAY								
	-BT3		THERMOSTAT SIGNALLING HIGH TEMPERATURE ALARM		-KFM		MICROPROCESSORS								
	-BU		BUCHHOLZ RELAY		-KFN		ANTIPUMPING RELAYS								
	-BX1		UNIT WITH SENSORS FOR DETECTION OF INTERNAL ARCING		-KFO		OPENING RELAYS OR CONTACTORS								
E	-BX2		ADDITIONAL CURRENT SENSING UNIT FOR DETECTION OF INTERNAL ARCING		-KFP		PROGRAMMABLE LOGIC CONTROLLERS (PLC)		E						
	-BZ,-BZ4		MULTIFUNCTION SENSORS		-KFR		RECLOSING RELAYS								
	-BZ1		COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L1		-KFS		SYNCHRONIZING DEVICES								
	-BZ2		COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L2		-KFT		AUXILIARY TIME RELAYS, DELAY ELEMENTS								
	-BZ3		COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L3		-KFZ		CONTROLLERS								
	-MBC		CLOSING RELEASE OF CIRCUIT-BREAKER		-MAD		MOTOR FOR ELECTRICAL OPERATION OF DISCONNECTOR -QBD								
	-MAD1		MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD1		-MAD2		MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD2								
	-MAD2		MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD2		-MAD9		MOTOR FOR ELECTRICAL OPERATION OF SWITCH-DISCONNECTOR -QBS								
	-MAD9		MOTOR FOR ELECTRICAL OPERATION OF SWITCH-DISCONNECTOR -QBS												
	F					Date Prep.	15.11.2018	PROJECT TITLE		PT. ABB Sakti Industri	REFERENCE DESIGNATIONS SLD & GA Drawing MVD-29 & MVD-30 20kV Centralized Panel Room WICA		Scale		= H00
1		Issued for Approval	18.12.2018	YR	Prepared	YR	Customer						PT Multimas Nabati Asahan	& GAD	
0		First Issued	19.11.2018	YR	Checked	AW	Drawing Nbr.							Sh. 5	
Ind.		Revision	Date	Name	Approved	AI	Based on:	BE318042		Customer Doc. Nbr.		Cont. 23			
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A	-MAE	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE	-QBH	MANUAL CIRCUIT-BREAKERS
	-MAE1	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2	-QBL	LINKS
B	-MAE8	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE SUPPLIED TOGETHER WITH SWITCH-DISCONNECTOR -QBS	-QBM	MINIATURE SWITCH-DISCONNECTORS
	-MAS	MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING	-QBS	SWITCH-DISCONNECTORS
C	-MAT	MOTOR FOR ELECTRICAL OPERATION OF TRUCK RACKING-IN/OUT	-QBT	TRUCK
	-MBC	CLOSING RELEASE OF CIRCUIT-BREAKER	-QCA	EARTHING SWITCH FOR ELIMINATION OF INTERNAL ARCING
D	-MBC4	CLOSING RELEASE OF SWITCH-DISCONNECTOR -QBS	-QCE	EARTHING SWITCH
	-MBO1	FIRST OPENING RELEASE OF CIRCUIT-BREAKER	-QCE1	EARTHING SWITCH OF BUSBARS (SINGLE BUSBAR SYSTEM) OR OF FRONT BUSBARS (DOUBLE BUSBAR SYSTEM)
E	-MBO2	SECOND OPENING RELEASE OF CIRCUIT-BREAKER	-QCE2	EARTHING SWITCH OF REAR BUSBARS (DOUBLE BUSBAR SYSTEM)
	-MBO3	OPENING SOLENOID FOR OVERCURRENT RELEASE OF CIRCUIT-BREAKER OR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER	-QQ	CLUTCHES
F	-MBO4	OPENING RELEASE OF SWITCH-DISCONNECTOR -QBS	-RAA	ANTIFERRORESONANCE RESISTOR
	-MBU	UNDERVOLTAGE RELEASE OF CIRCUIT-BREAKER	-RAD	DIODES
G	-PFB	BLUE SIGNAL LAMPS	-RAI	INDUCTORS
	-PFF	FLAG RELAYS	-RAR	RESISTORS
H	-PFG	GREEN SIGNAL LAMPS	-RF	FILTERS
	-PFR	RED SIGNAL LAMPS	-RLE1	ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING
I	-PFS	SHORT CIRCUIT INDICATORS	-RLE2	ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT
	-PFV	VOLTAGE INDICATORS	-RLE3	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE
J	-PFV1	VOLTAGE INDICATORS ON FEEDER SIDE	-RLE4	ELECTROMECHANICAL LOCK PREVENTING THE DOOR OPENING OPERATION
	-PFV2	VOLTAGE INDICATORS ON BUSBAR SIDE	-RLE5	ELECTROMECHANICAL LOCK PREVENTING OPERATION OF DISCONNECTOR -QBD
K	-PFW	WHITE SIGNAL LAMPS	-RLE6	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD1
	-PFX	CROSS INDICATORS, ELECTROMECHANICAL INDICATORS	-RLE7	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD2
L	-PFY	YELLOW SIGNAL LAMPS	-RLE8	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2
	-PGA	AMMETERS	-RLE9	ELECTROMECHANICAL LOCK PREVENTING OPERATION OF SWITCH-DISCONNECTOR -QBS
M	-PGC	COUNTERS	-RLE11	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD4
	-PGF	FREQUENCYMETERS	-RLE12	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD5
N	-PGH	HOURMETERS	-SFA	AMMETRIC SWITCHES
	-PGI	PROTECTION AND CONTROL UNIT: HUMAN MACHINE INTERFACE	-SFC	CONTROL SWITCHES, CLOSING PUSH-BUTTONS
O	-PGI1	HUMAN MACHINE INTERFACE OF ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON CIRCUIT-BREAKER)	-SFL	LOCKING CONTACTS
	-PGI2	HUMAN MACHINE INTERFACE FOR ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON SWITCHGEAR)	-SFO	OPENING PUSH-BUTTONS
P	-PGJ	ACTIVE ENERGY METERS	-SFR	RESET PUSH-BUTTONS
	-PGK	REACTIVE ENERGY METERS	-SFS	SELECTOR SWITCHES
Q	-PGM	MULTIFUNCTION INDICATORS	-SFT	TEST PUSH-BUTTONS
	-PGP	POWER-FACTOR METERS	-SFU	UNLOCKING PUSH-BUTTONS
R	-PGQ	VARMETERS	-SFV	VOLTMETRIC SWITCHES
	-PGS	SYNCHRONOSCOPES	-TA	POWER TRANSFORMERS
S	-PGV	VOLTMETERS	-TB	CONVERTER
	-PGW	WATTMETERS	-TB1	RECTIFIER FOR FIRST OPENING RELEASE
T	-PJ	ACOUSTICAL SIGNAL DEVICES (BELLS, SIRENS)	-TB2	RECTIFIER FOR SECOND OPENING RELEASE
	-QAB	CIRCUIT-BREAKERS	-TB3	RECTIFIER FOR CLOSING RELEASE
U	-QAC	CONTACTORS (FOR POWER)	-TB4	RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING
	-QBD	DISCONNECTORS	-TB5	RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT
V	-QBD1	FRONT BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)	-TB6	RECTIFIER FOR UNDERVOLTAGE RELEASE
	-QBD2	REAR BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)	-TB7	ERECTIFIER FOR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER
W	-QBD3	FEEDER DISCONNECTORS (DOUBLE BUSBAR SYSTEM)		
	-QBD4	TWO-WAY DISCONNECTOR		
X	-QBD5	TWO-WAY DISCONNECTOR ON BUSBAR RISER		

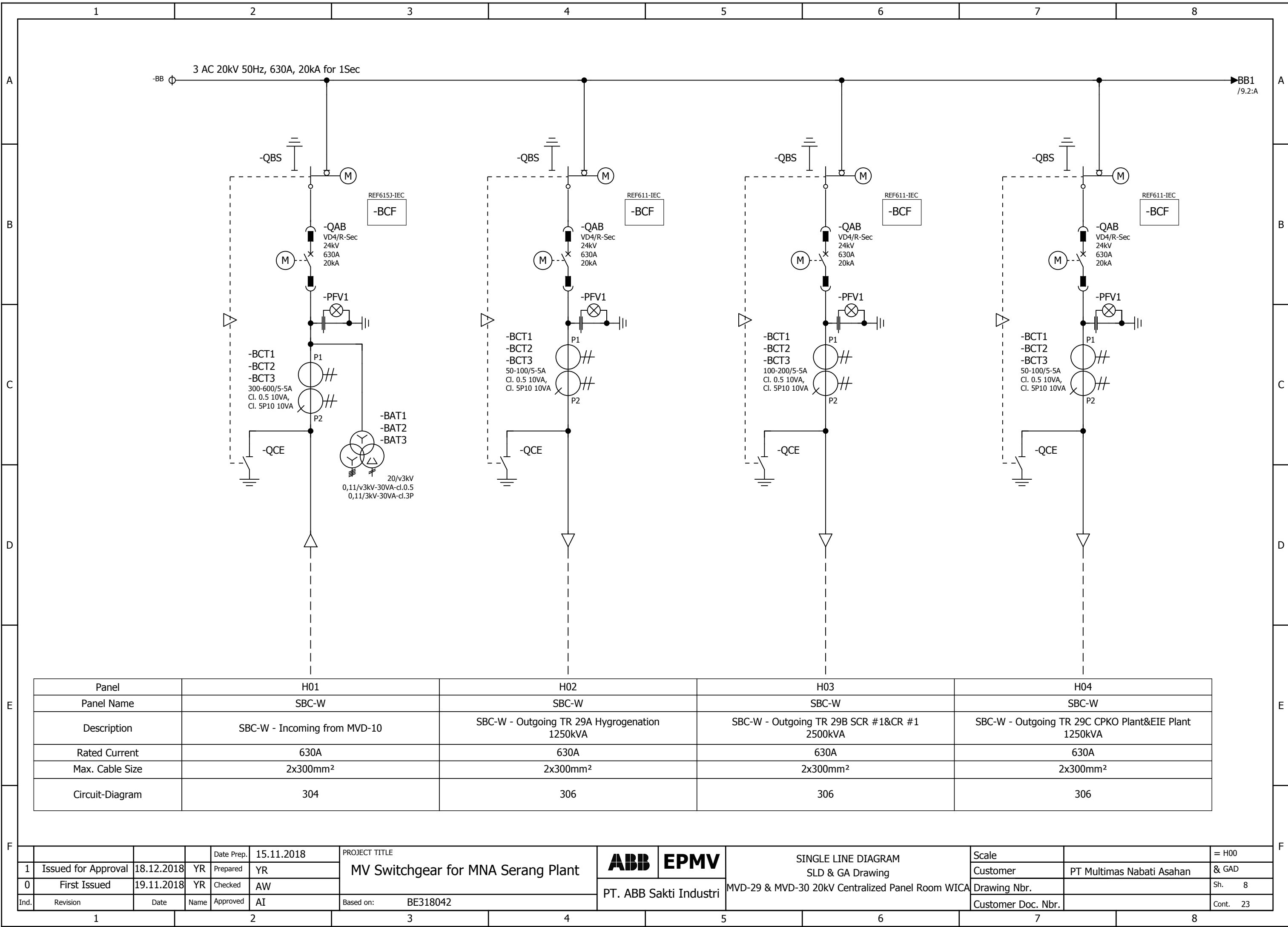
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1	Issued for Approval	18.12.2018	YR	Prepared	YR	MV Switchgear for MNA Serang Plant					Customer	PT Multimas Nabati Asahan	& GAD
0	First Issued	19.11.2018	YR	Checked	AW			PT. ABB Sakti Industri	MVD-29 & MVD-30 20kV Centralized Panel Room WICA	Drawing Nbr.		Sh. 6	
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042				Customer Doc. Nbr.		Cont. 23	

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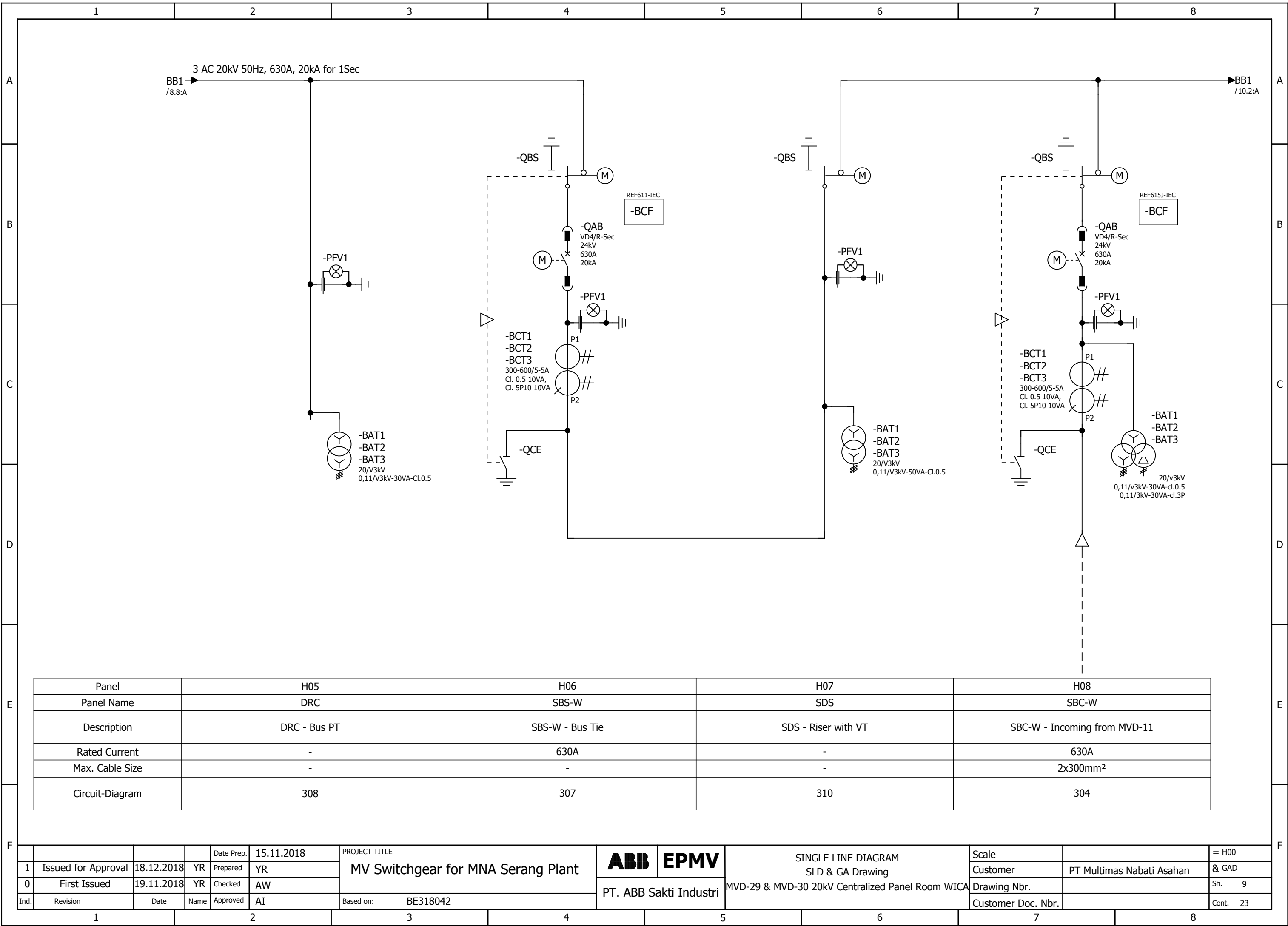
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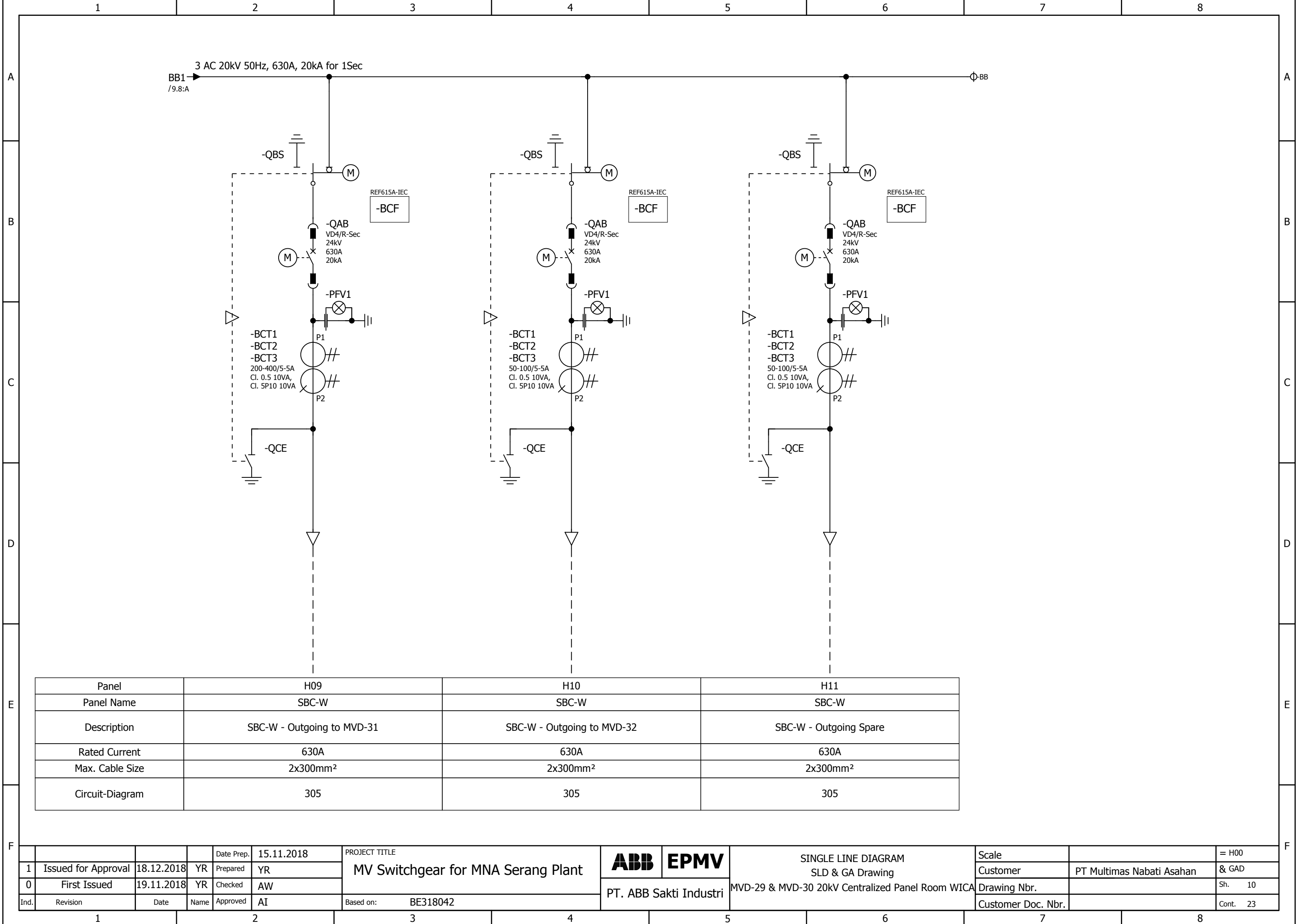
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A	-TFA	ACTIVE POWER TRANSDUCERS						-XD11	TERMINAL BLOCK FOR INTERCONNECTIONS OF CONTROL CIRCUITS						A									
	-TFC	CURRENT TRANSDUCERS						-XD12	TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING															
	-TFF	FREQUENCY TRANSDUCERS						-XD13	TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR SWITCH-DISCONNECTOR OPERATION															
	-TFJ	ACTIVE ENERGY TRANSDUCERS						-XD14	TERMINAL BLOCK FOR INTERCONNECTIONS OF A.C. AUXILIARY CIRCUITS															
	-TFK	REACTIVE ENERGY TRANSDUCERS						-XD16	TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS) OF VOLTAGE CIRCUITS															
	-TFM	MULTIFUNCTION TRANSDUCERS																						
B	-TFP	POWER-FACTOR TRANSDUCERS						-XD17	TERMINAL BLOCK FOR INTERCONNECTION OF MOD-BUS						B									
	-TFQ	REACTIVE POWER TRANSDUCERS						-XD18	TERMINAL BLOCK FOR INTERCONNECTIONS OF CURRENT CIRCUITS															
	-TFS	SIGNAL CONVERTERS						-XD19	TERMINAL BLOCK FOR SPECIAL USE															
	-TFV	VOLTAGE TRANSDUCERS						-XDJ	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF DISCONNECTORS -QBD															
	-WA	BUSBARS						-XDJ1	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR -QBS															
	-WBC	POWER CABLES						-XDM	SEALABLE TERMINAL BLOCK FOR MEASUREMENT															
C	-WE	EARTHING CABLES FOR L.V. INSTRUMENT COMPARTMENT.						-XDS	CURRENT SOCKETS						C									
	-WGC	CONTROL CABLES						-XDT	TERMINAL BLOCK FOR POSITION CONTACTS OF TRUCK AND OF CONNECTOR -XDB															
	-WGS	COMMUNICATION CABLES (SHIELDED TWISTED PAIRS)						-XDU	CONNECTOR FOR ISOLATION OF CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK															
	-WH	OPTICAL FIBRES						-XDU1	TERMINAL BLOCK FOR CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK															
	-XDA	TERMINAL BLOCK FOR CIRCUITS OF CURRENT TRANSFORMERS						-XDV	TERMINAL BLOCK FOR CIRCUITS OF VOLTAGE TRANSFORMERS															
	-XDA1	CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L1						-XDV1	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L1															
D	-XDA2	CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L2						-XDV2	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L2						D									
	-XDA3	CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L3						-XDV3	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L3															
	-XDA5	CONNECTOR FOR CIRCUITS OF NEUTRAL (RESIDUAL) CURRENT TRANSFORMER						-XDV4	CONNECTOR FOR CIRCUITS OF ANTIFERRORESONANCE RESISTOR															
	-XDB	CONNECTOR FOR ISOLATION OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK) AUXILIARY CIRCUITS						-XDX	SUPPORT TERMINAL BLOCK															
	-XDB1	CONNECTOR FOR AUXILIARY CIRCUITS OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK)						-XE1	EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, LEFT SIDE															
	-XDB2	CONNECTOR FOR ADDITIONAL AUXILIARY CIRCUITS OF CIRCUIT-BREAKER						-XE2	EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, RIGHT SIDE															
E	-XDB10....-XDB89	CONNECTOR FOR CIRCUIT-BREAKER INTERNAL CIRCUITS						-XE5	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, VOLTAGE INDICATOR SIDE						E									
	-XDB7	CONNECTOR FOR TRUCK POSITION CONTACTS						-XE6	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, REAR BOTTOM SIDE															
	-XDB91	CONNECTOR FOR OPENING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR						-XE7	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, LEFT SIDE															
	-XDB92	CONNECTOR FOR CLOSING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR						-XE8	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, FRONT BOTTOM SIDE															
	-XDB93	CONNECTOR FOR POSITION SWITCHES OF SWITCH-DISCONNECTOR						-XE9	EARTHING TERMINAL BLOCK FOR TRANSFORMERS IN BUSBAR COMPARTMENT															
	-XDB95	CONNECTOR FOR CIRCUITS OF EARTHING SWITCH LOCKING MAGNET																						
F	-XDC	CUSTOMER TERMINAL BLOCK													F									
	-XDC1	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF CIRCUIT BREAKER																						
	-XDC2	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR																						
	-XDC3	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF PROTECTION RELAY AND MULTIFUNCTION UNIT																						
	-XDC4	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF VOLTAGE TRANSFORMERS																						
	-XDC5	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF MINIATURE CIRCUIT BREAKER																						
F	-XDE	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE													F									
	-XDE1	TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE																						
	-XDE2	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2																						
	-XDE3	TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2																						
	-XDF	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF FORCED COOLING VENTILATORS																						
	-XDF1	CONNECTOR FOR ISOLATION OF CIRCUITS OF FRONT VENTILATOR																						
F	-XDF2	CONNECTOR FOR REAR VENTILATOR CIRCUITS													F									
	-XDF3	CONNECTOR FOR FRONT VENTILATOR CIRCUITS																						
	-XDH	TERMINAL BLOCK FOR VARIOUS CIRCUITS (HEATERS, POSITION SWITCHES DETECTING INTERNAL ARCING, ETC.)																						
	-XDI	TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS)																						
1		2		3		4		5		6		7		8										
				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant		ABB	EPMV	REFERENCE DESIGNATIONS SLD & GA Drawing		Scale		= H00										
1	Issued for Approval	18.12.2018	YR	Prepared	YR							Customer	PT Multimas Nabati Asahan	& GAD										
0	First Issued	19.11.2018	YR	Checked	AW							Drawing Nbr.		Sh. 7										
Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318042	PT. ABB Sakti Industri		MVD-29 & MVD-30 20kV Centralized Panel Room WICA		Customer Doc. Nbr.		Cont. 23										
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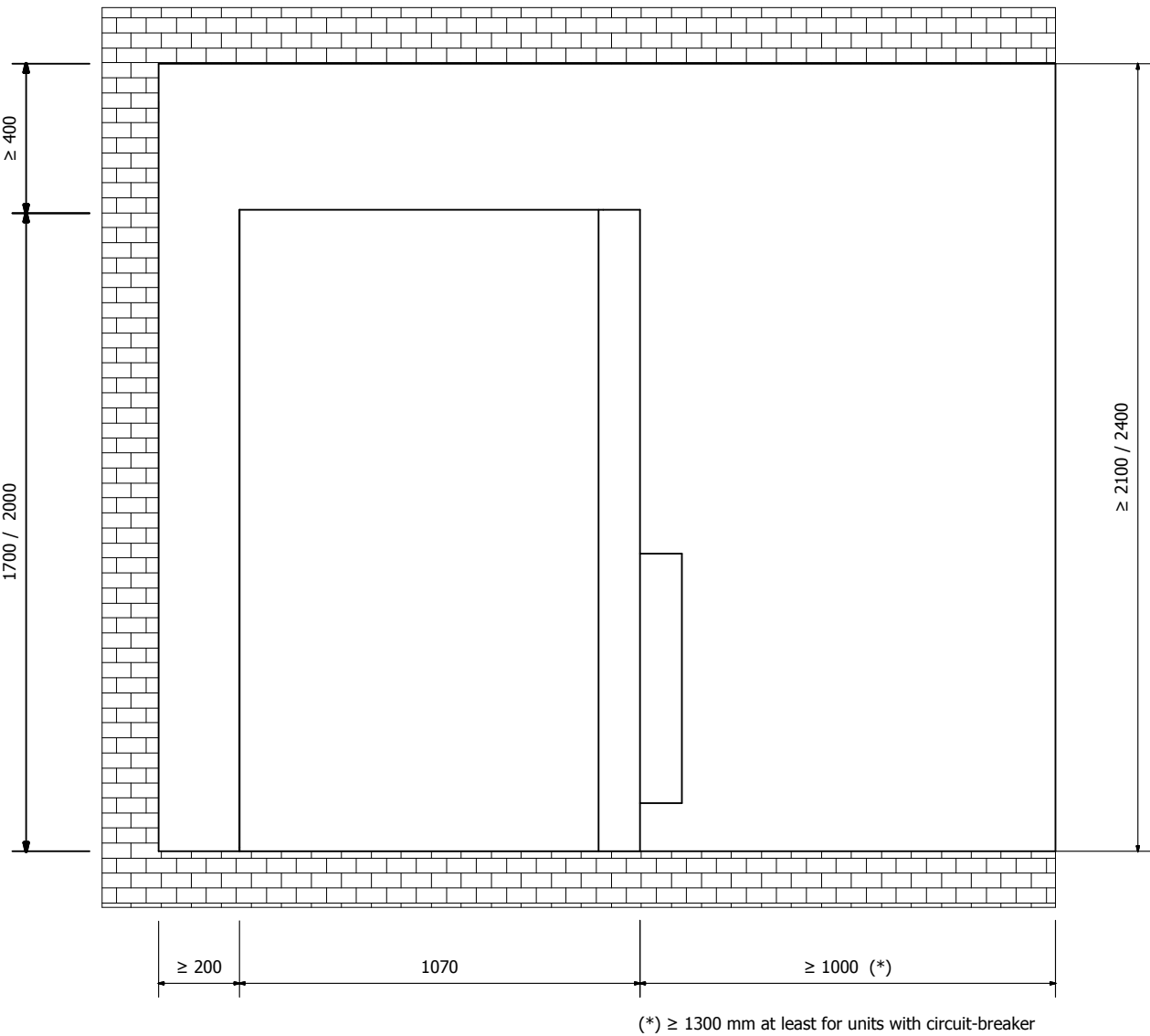
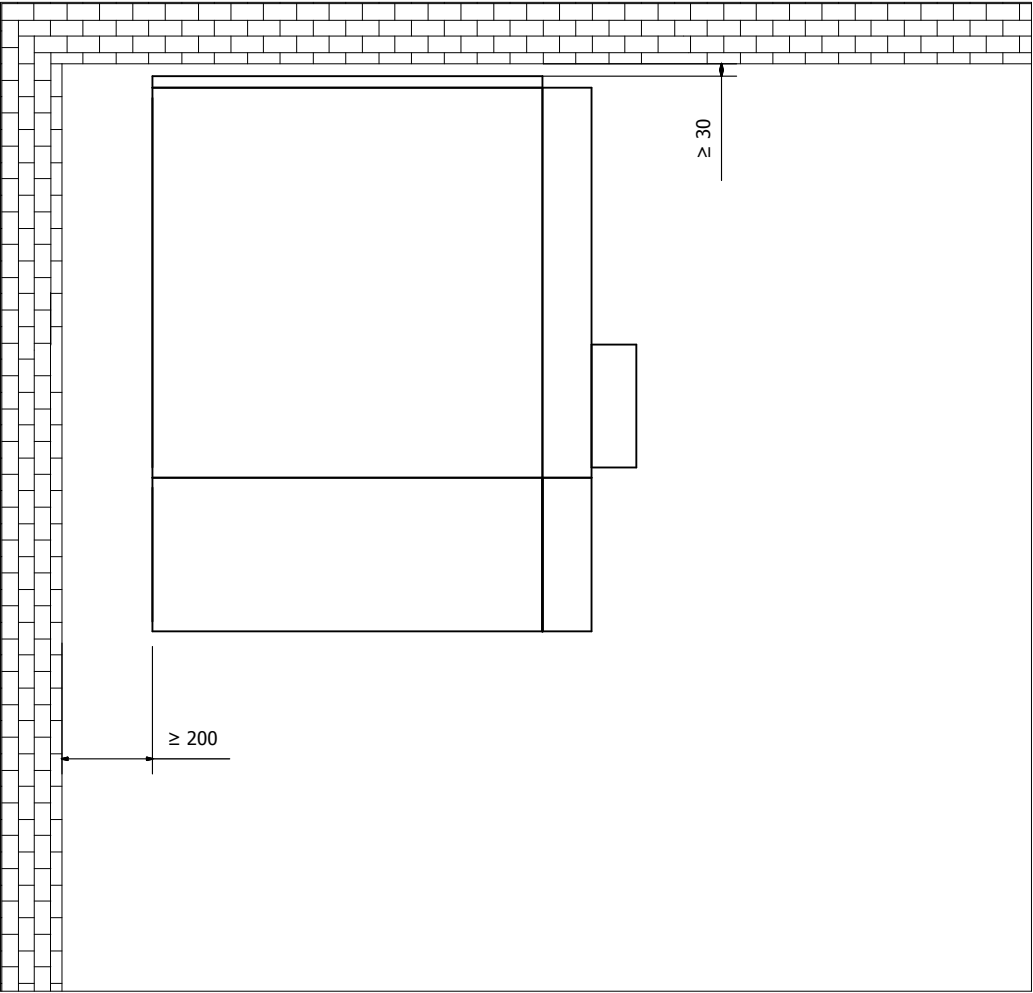


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RULES FOR INSTALLATION IN THE ROOM

MINIMUM DISTANCES TO BE RESPECTED DURING INSTALLATION



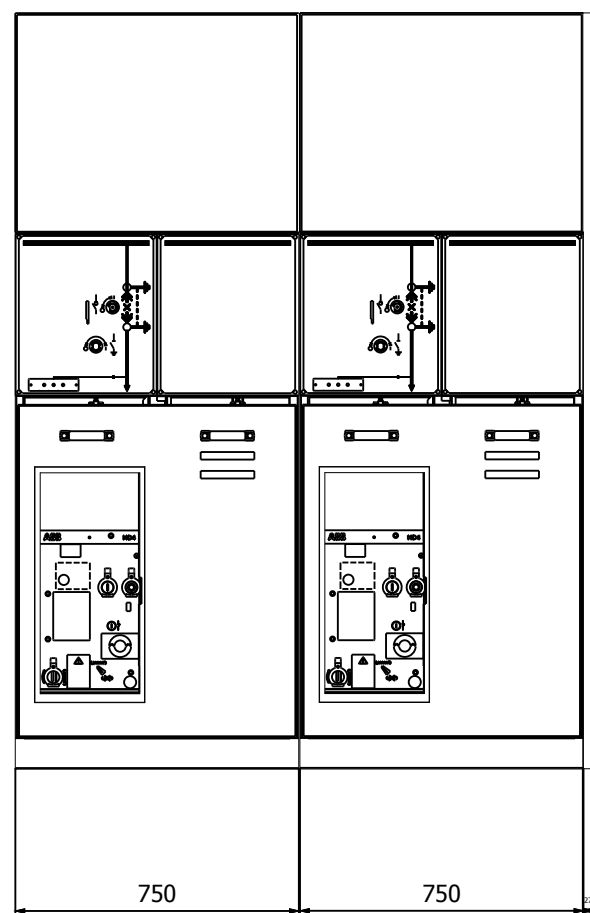
- IAC A-F SOLUTION UP TO 16kA 1sec
- ATTENTION : NO ACCESS TO REAR AND LATERAL OF THE SWG ROOM WHILE SWG IS IN SERVICE

				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	INSTALLATION RULES SLD & GA Drawing MVD-29 & MVD-30 20kV Centralized Panel Room WICA	Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR					Customer	PT Multimas Nabati Asahan	& GAD
0	First Issued	19.11.2018	YR	Checked	AW		PT. ABB Sakti Industri	Drawing Nbr.			Sh. 11	
Ind.	Revision	Date	Name	Approved	AI			Based on: BE318042		Customer Doc. Nbr.		Cont. 23

Panel	H01	H02	H03	H04	H05	H06	H07	H08	H09
Panel Type	SBC-W	SBC-W	SBC-W	SBC-W	DRC	SBS-W	SDS	SBC-W	SBC-W
Description	SBC-W - Incoming	SBC-W - Outgoing TR 1600kVA	SBC-W - Outgoing TR 1600kVA	SBC-W - Outgoing TR 1600kVA	DRC - Bus PT	SBS-W - Bus Tie	SDS - Riser with VT	SBC-W - Incoming	SBC-W - Outgoing

				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	FRONT VIEW & GA SLD & GA Drawing	Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR					Customer	PT Multimas Nabati Asahan	& GAD
0	First Issued	19.11.2018	YR	Checked	AW		PT. ABB Sakti Industri	MVD-29 & MVD-30 20kV Centralized Panel Room WICA	Drawing Nbr.		Sh. 12	
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042			Customer Doc. Nbr.		Cont. 23	

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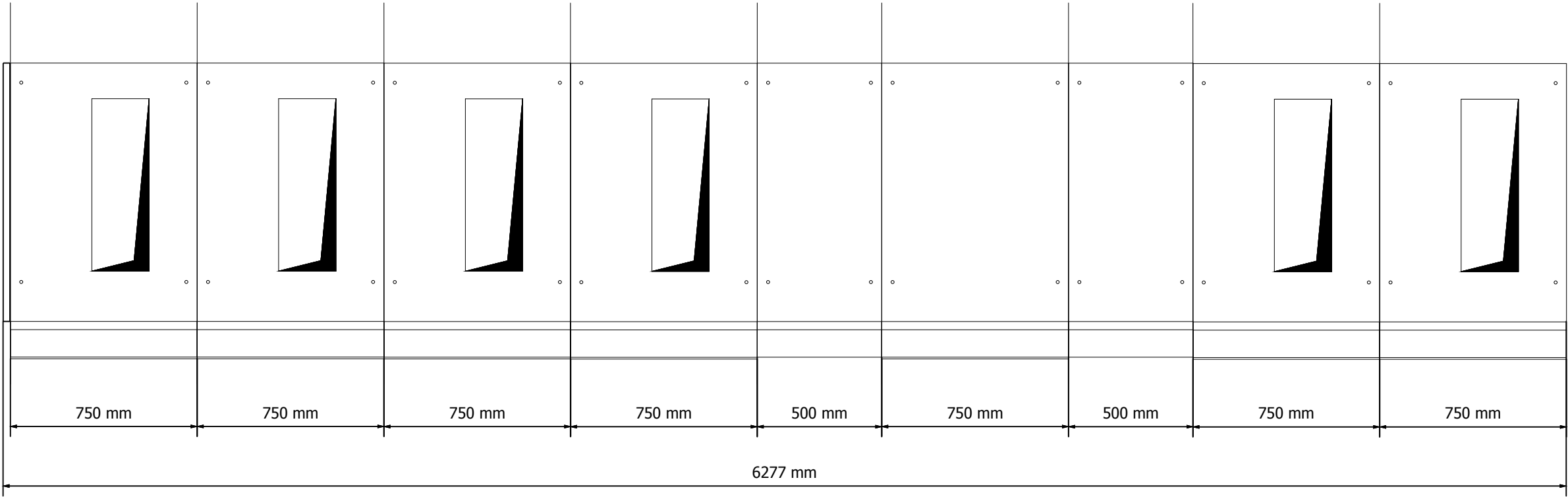


Panel	H10	H11
Panel Type	SBC-W	SBC-W
Description	SBC-W - Outgoing	SBC-W - Outgoing

				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	 	FRONT VIEW & GA SLD & GA Drawing	MVD-29 & MVD-30 20kV Centralized Panel Room WICA	Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR					Customer	PT Multimas Nabati Asahan	& GAD
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FOUNDATION FRAMES

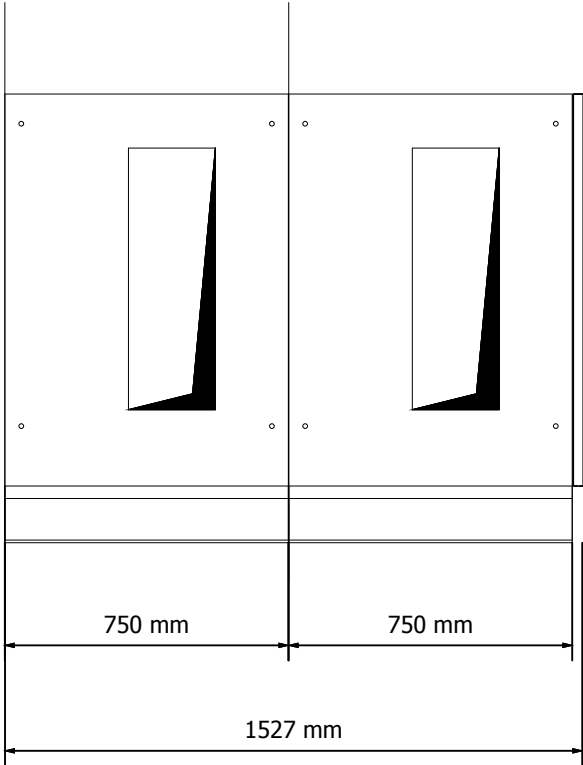


Panel	H01	H02	H03	H04	H05	H06	H07	H08	H09
Panel Type	SBC-W	SBC-W	SBC-W	SBC-W	DRC	SBS-W	SDS	SBC-W	SBC-W
Description	SBC-W - Incoming	SBC-W - Outgoing TR 1600kVA	SBC-W - Outgoing TR 1600kVA	SBC-W - Outgoing TR 1600kVA	DRC - Bus PT	SBS-W - Bus Tie	SDS - Riser with VT	SBC-W - Incoming	SBC-W - Outgoing

				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	FOUNDATION FRAMES SLD & GA Drawing	Scale		= H00
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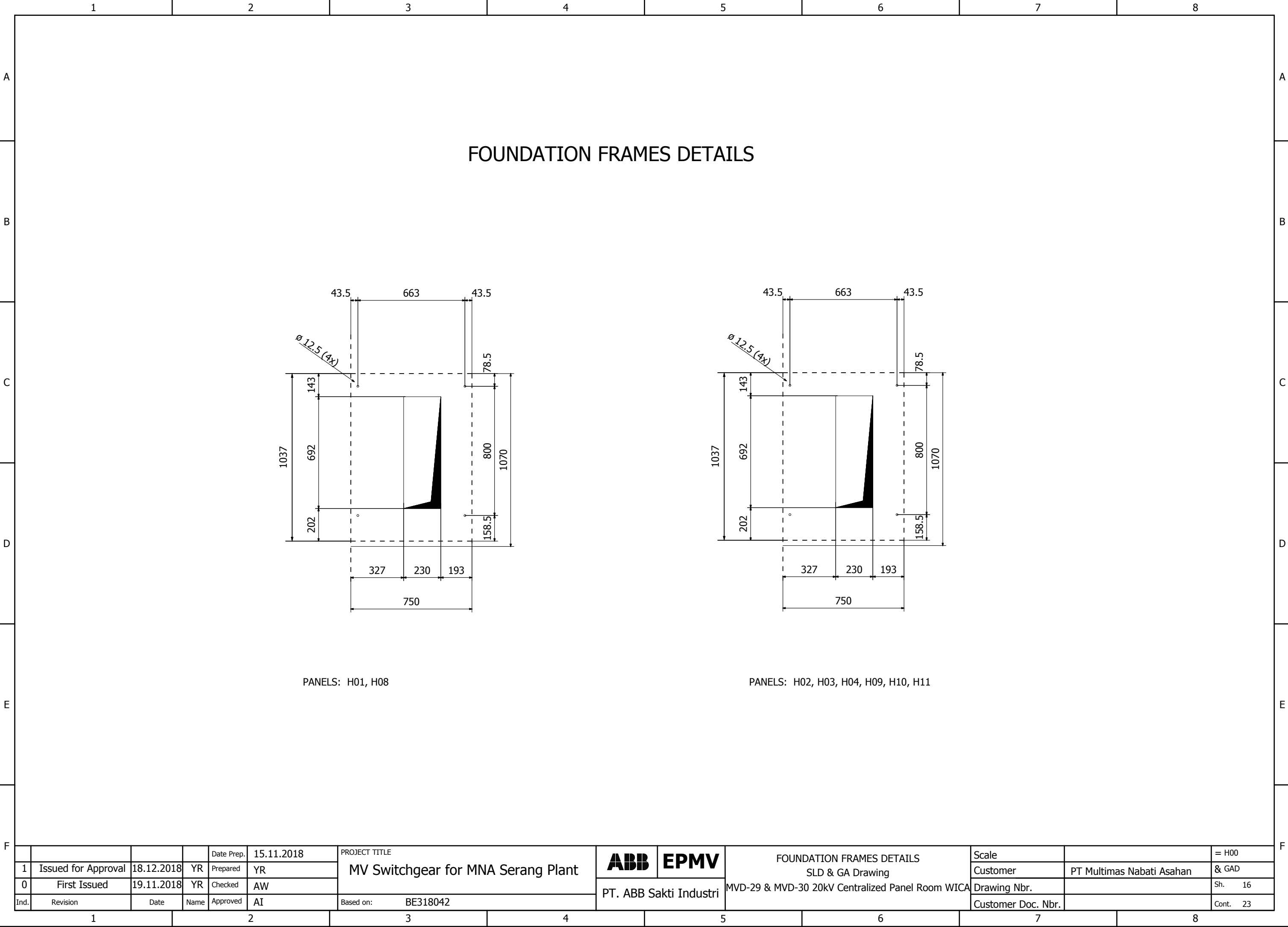
FOUNDATION FRAMES



Panel	H10	H11
Panel Type	SBC-W	SBC-W
Description	SBC-W - Outgoing	SBC-W - Outgoing

				Date Prep.	15.11.2018	PROJECT TITLE			Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR	MV Switchgear for MNA Serang Plant			Customer	PT Multimas Nabati Asahan	& GAD
0	First Issued	19.11.2018	YR	Checked	AW				Drawing Nbr.		Sh. 15
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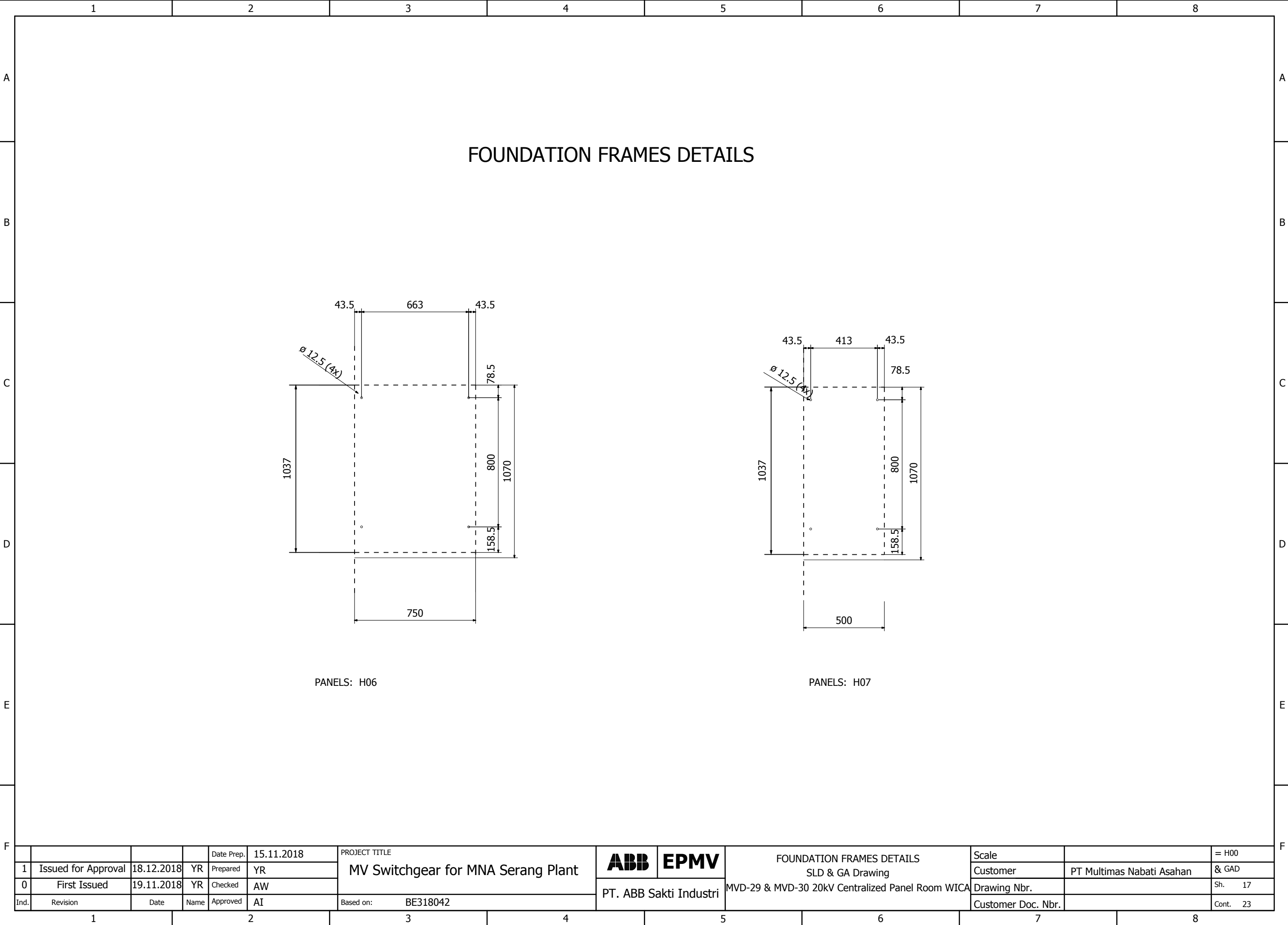
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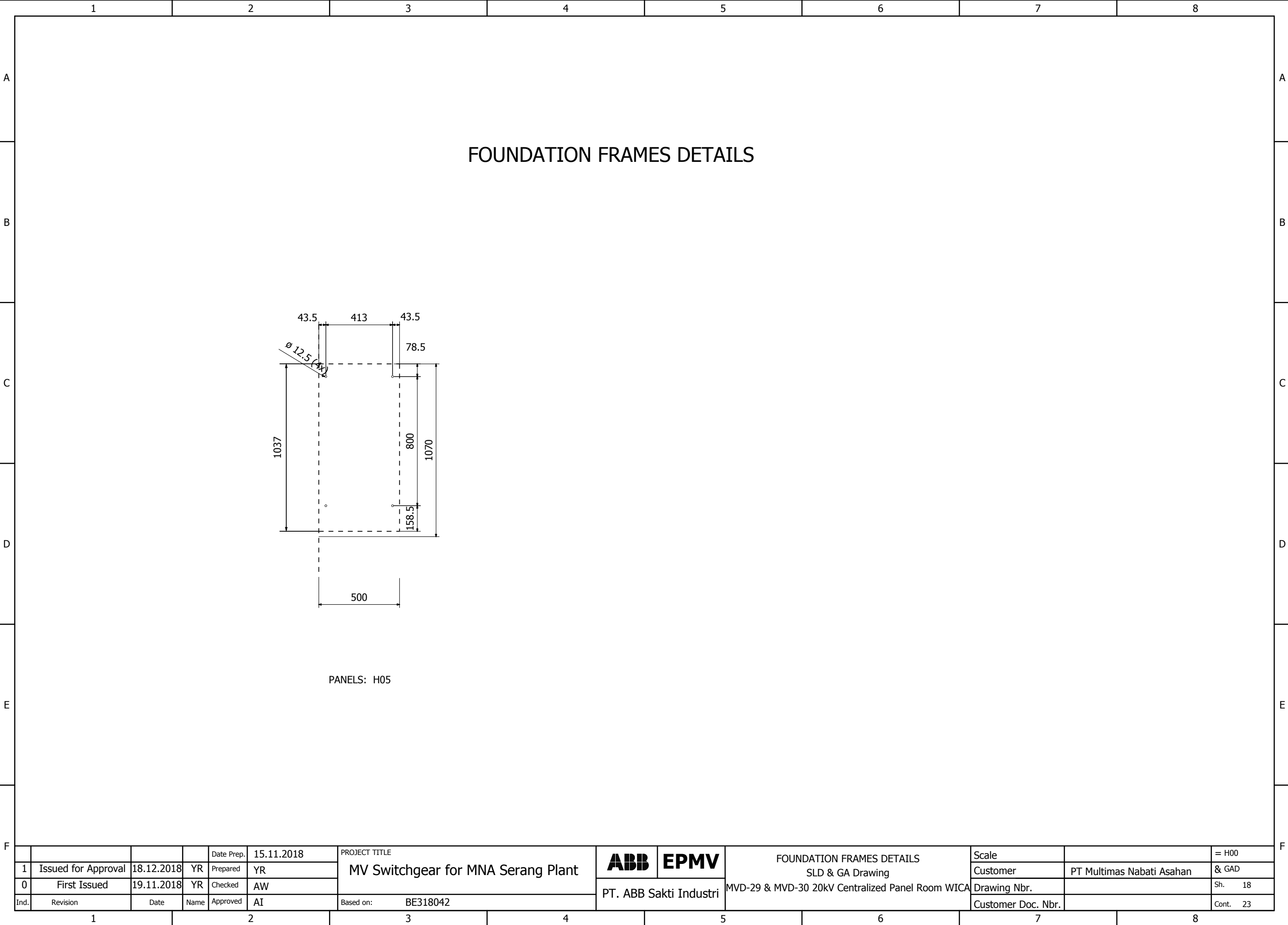
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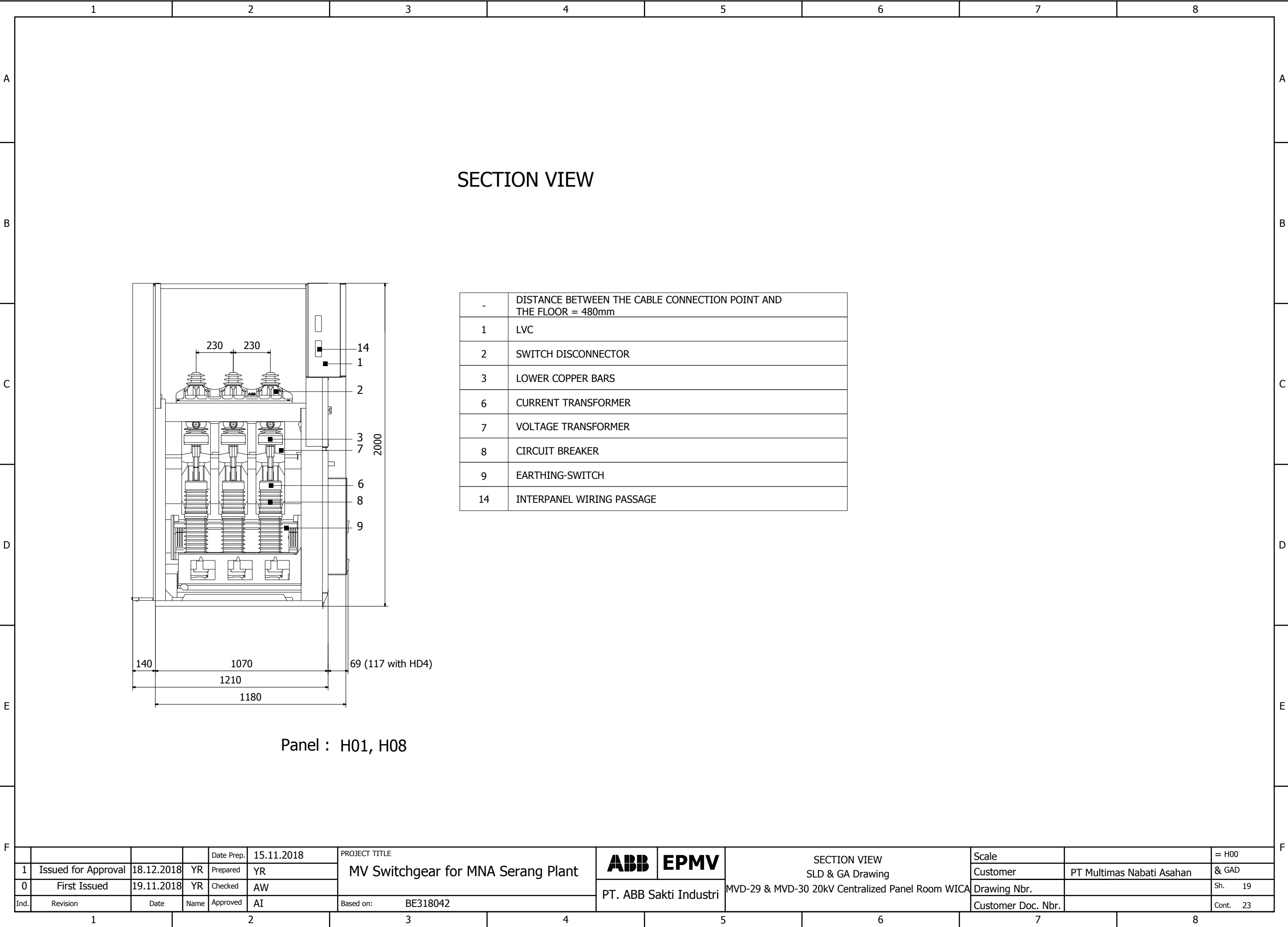
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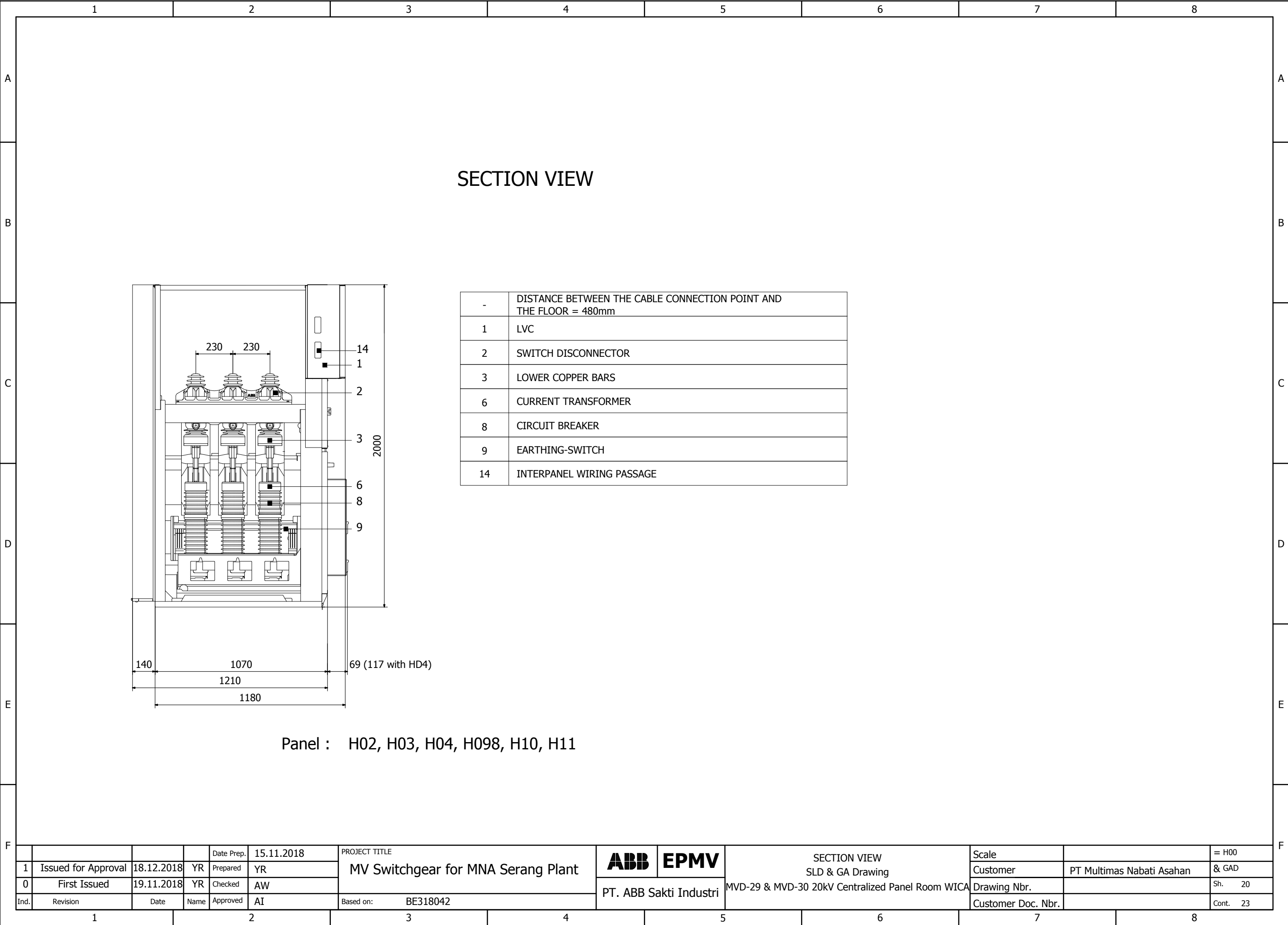
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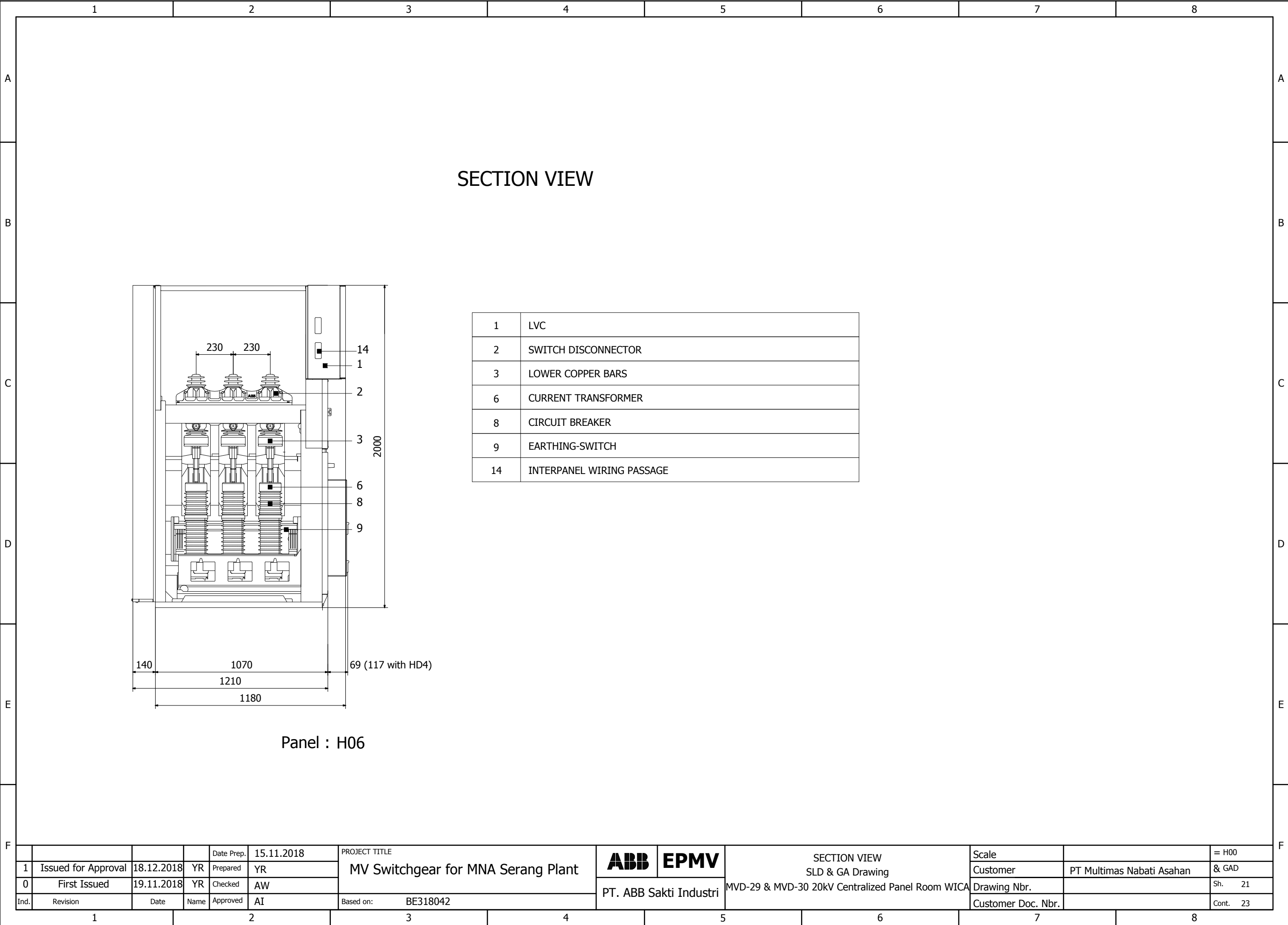
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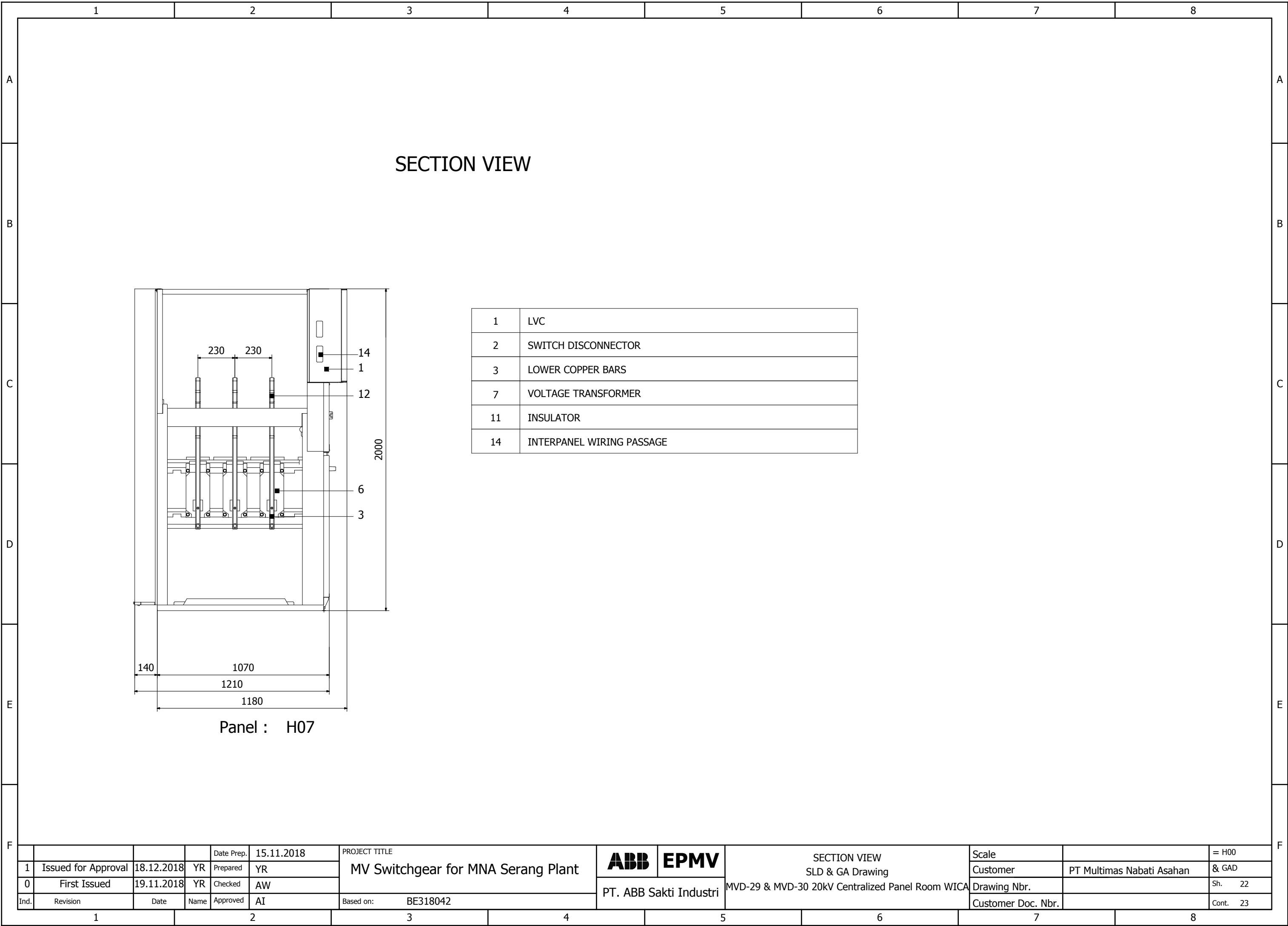
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				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	SECTION VIEW SLD & GA Drawing MVD-29 & MVD-30 20kV Centralized Panel Room WICA	Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR					Customer	PT Multimas Nabati Asahan	& GAD
0	First Issued	19.11.2018	YR	Checked	AW		PT. ABB Sakti Industri	Drawing Nbr.			Sh. 22	
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042				Customer Doc. Nbr.		Cont. 23

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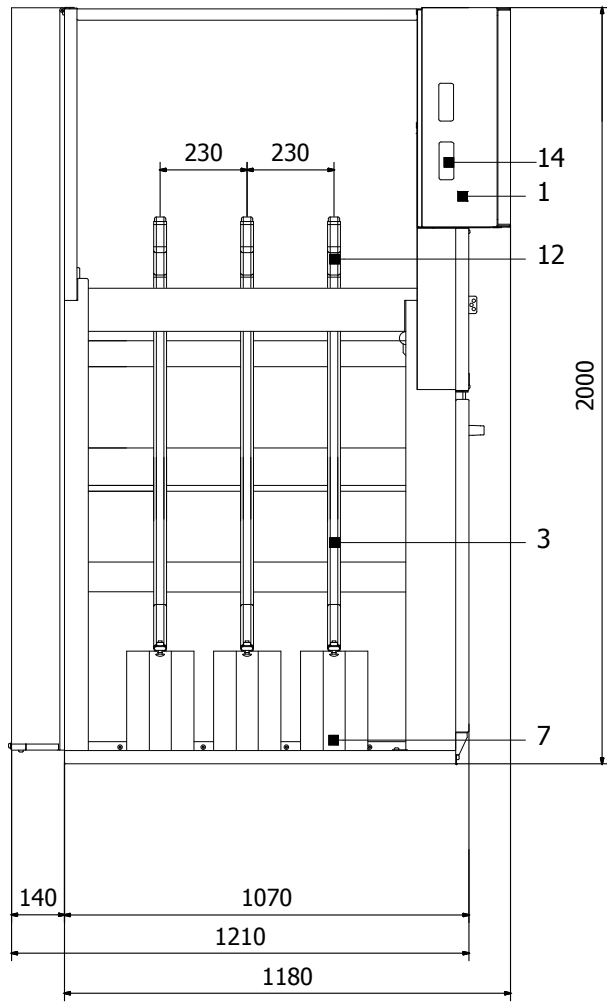
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SECTION VIEW



-	DISTANCE BETWEEN THE CABLE CONNECTION POINT AND THE FLOOR = 670mm
1	LVC
3	LOWER COPPER BARS
6	VOLTAGE TRANSFORMER
12	TOP COPPER BARS
14	INTERPANEL WIRING PASSAGE

Panel : H05

				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant		ABB EPMV	SECTION VIEW SLD & GA Drawing MVD-29 & MVD-30 20kV Centralized Panel Room WICA		Scale		= H00
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